Intelligent agents in a Smart Home

An Intelligent agent observes its environment through Sensors and acts on it through Sensors devices. In a Smart Home, Sensors might detect motion, temperature, light and time; devices such as lights, thermostats, & door locks carryout the agents action.

Simple Reflex Agent:

Acts only on the current input using condition - action rules. For example, if motion is detected after Sunset, turn on the hallway light.

Model-based Agent:

Keeps an internal model of home and uses it to interpret current Sensor readings. This helps when information is incomplete, though the model must be kept accurate.

Goal-based agent:

chooses actions that help achieve a stated goal, such as keeping the living room within a comfortable temperature range or ensuring the doors are locked at night.

utility-based agent :

Scores possible outcomes according to factors such as comfort, energy use, & security then selects an action with highest overal value.

A hybrid approach would work best. Simple Reflex rules can handle urgent events such as smoke detection or an open door at night. A model based component can track occupancy and environmental changes. Goal-based planning can support routines, while a utility-based component balances comfort, energy efficiency and security. Learning from user preferences can improve the model and utility scores over time. Safety rules should remain fixed for critical situations.

an unknown maze

Uniform cost Search (UCS): Expands the node with the lowest Path cost $g(n)$

Advantages: Complete and optimal, even when Step costs vary (turns, rough terrain).

Disadvantages: Time and Space are $O(b^{1+(C^*/\varepsilon)})$. It Stores the whole Frontier, so memory grows fast, and it Explores in all direction.

Iterative Deepening Search (IDS):

Runs depth-limited DFS with limits 0, 1, 2... until the exit is found.

Advantages: Complete and optimal when Step costs are uniform.

Disadvantages: It regenerates upper-level nodes repeatedly (small overhead), ignores Step costs and needs Cycle Checking.

2. Robot in an unknown maze

Uniform cost Search (UCS): Expands the node with the Lowest Path cost $g(n)$

Advantages: Complete and optimal, even when Step Costs vary (turns, rough terrain).

Disadvantages: Time and Space are $O(b^{1+C^*/\varepsilon})$. It Stores the whole Frontier, so memory grows fast, and it Explores in all direction.

Iterative Deepening Search (IDS):

Runs depth-limited DFS with limits 0, 1, 2... until the exit is found.

Advantages: Complete and optimal when Step costs are uniform.

Disadvantages: It regenerates upper-level nodes repeatedly (small overhead), ignores Step costs and needs Cycle Checkip.

Relation to agent Types :

Simple reflex : fails, as it has no memory and loops in the maze.

Model-based : needed, since the maze is unknown. It builds a map & remembers visited cells.

Goal based : Searches for a Path to the exit.

Utility based : Useful when moves differ in cost (energy, time, risk).

Recommended combination : a model based, goal based agent with IDS, which is memory efficient and finds the shallowest exit. If move costs vary, use a utility based agent with UCS for the optimal least-cost Path, accepting hiher memory use.